

*Calculators may be used in this examination but must not be used to store text.
Calculators with the ability to store text should have their memories deleted prior
to the start of the examination.*

UNIVERSITY OF BIRMINGHAM

School of Civil Engineering

Level C

**04 23632
(CE1SAM)**

Statics and Mechanics

Summer Examinations 2012

Time Allowed: 3 hours

Answer **FIVE** questions only

All questions carry equal marks. The allocation of marks within each question is stated in the right-hand margin.

1. Determine the support reactions for the structure shown in Figure Q1. [20]

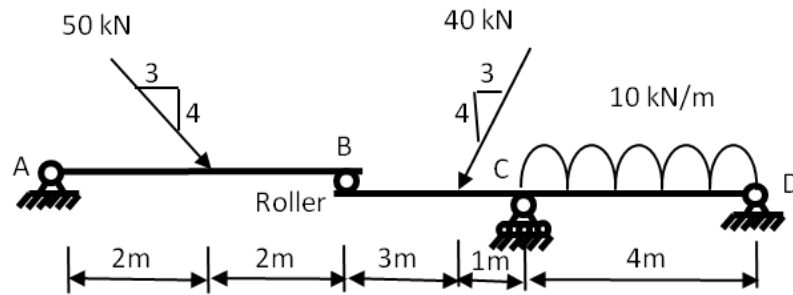


Figure Q1

2. With reference to Figure Q2,

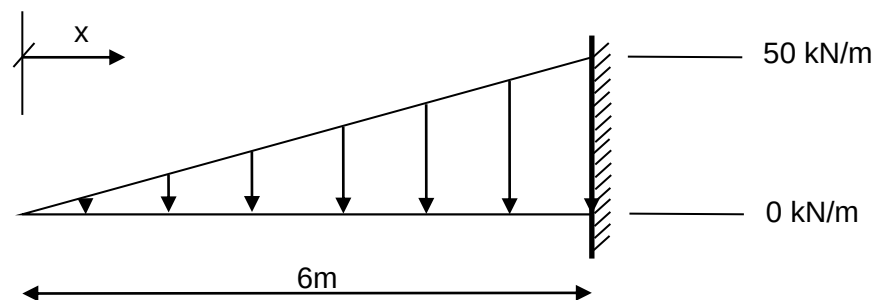


Figure Q2

- a. Show that the following expressions are valid:

$$m(x) = \frac{25}{18}x^3 \quad \& \quad \frac{m(x)}{F(x)} = \frac{x}{3},$$

where the symbols have their usual meaning. [8]

- b. Draw the shear force and bending moment diagrams noting the key values on each. [7]

- c. Show that the magnitude of the maximum deflection of the beam in Figure Q2 is [5]

$$\frac{5}{18} \frac{L^5}{EI}$$

3. The cylindrical brass post (diameter 50mm, length 0.5m and $E=105 \text{ kN/mm}^2$) is encased by an outer (circular) core of aluminium ($E = 70 \text{ kN/mm}^2$, length 0.5m). The thickness of the aluminium is 25mm, i.e., the total diameter of the combined post is 100mm. If a rigid cap is added to the top of the post and an axial load of 60 kN is applied show that force in the aluminium is twice the force in the brass. Hence determine the direct stress in each of the materials. [20]

4. Consider the shape illustrated in Figure Q4.

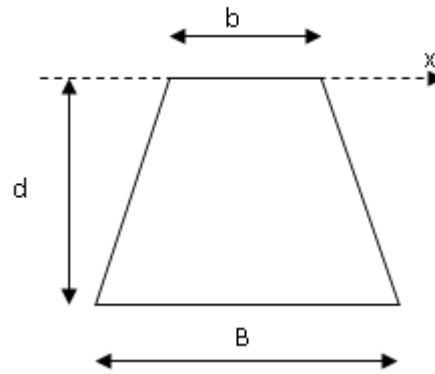


Figure Q4

- a. Show that the centroid of the shape is located at a distance of:
- $$\frac{d}{3} \frac{(2B+b)}{(3B-b)}$$
- from the x-axis and that the corresponding second moment of area of the shape about the x axis is:

$$\frac{d^3}{12} (3B + b)$$

[10]

- b. Hence, or otherwise show that the second moment of area of the shape about its centroid is given by:

$$\frac{d^3}{12(3B-b)} (17B^2 + 8Bb + b^2)$$

[10]

5. a. Consider the beam cross-section shown in Figure Q5a (all dimensions are in mm).

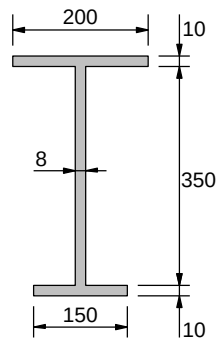


Figure Q5a

What is the magnitude of the plastic section modulus?

[10]

- b. The simply supported uniform beam shown in Figure Q5b has a fully-plastic moment M_p .

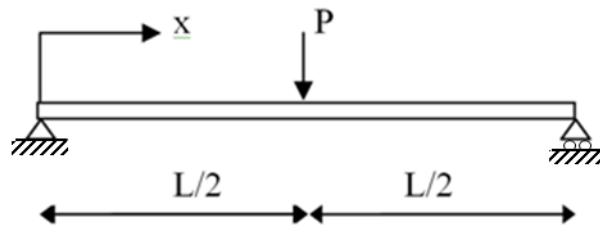


Figure Q5b

What is the magnitude of P required to cause collapse?

[10]

6. Question is irrelevant

[10]

[10]

7. Question is irrelevant

[20]